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Financial Voucher Analysis with LVMs and Financial LLMs

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***Abstract*—This work presents a novel computational framework for automated financial voucher analysis that combines large vision-language models (LVMs) and specialized financial large language models (LLMs) into a multi-stage, scalable pipeline. Leveraging transformer-based architectures, including a domain-specific 7 billion parameter financial language model, the system converts unstructured voucher data into structured representations to uncover meaningful financial patterns. The framework uses statistical methods like Z-score analysis and coefficient of variation to allow anomaly identification in three dimensions: amount-based, temporal, and pattern-based. Extraction and analysis processes are formalized through mathematical models and implemented algorithmically, with built-in fallback mechanisms to handle model uncertainty and ensure statistical validity. Our comparative evaluation demonstrates that specialized financial LLMs provide a 19.9% performance improvement over general-purpose LLMs, with particular gains in financial terminology recognition (+18.9%) and numerical accuracy (+13.2%). Evaluation on financial voucher data from different companies demonstrates the effectiveness of the framework in cash flow analysis, tax compliance, expense categorization, and anomaly detection, with a 15.7% reduction in false positives for anomaly identification. This methodology advances the state of automated financial document analysis, with significant implications for auditing, compliance, and business intelligence.**

***Index Terms*—Finance, LLMs, LVMs,**

I. Introduction

Financial document analysis remains a critical yet labor-intensive aspect of accounting, auditing, and financial compliance. Organizations process hundreds or thousands of vouchers annually, each containing valuable financial data that must be accurately extracted, categorized, and analyzed to ensure regulatory compliance and support business decision-making. Traditional manual processing of these documents is time-consuming, error-prone, and fails to scale effectively with increasing document volumes. Moreover, manual analysis often struggles to identify subtle patterns, anomalies, or inconsistencies that may indicate errors, inefficiencies, or potential compliance issues.

Recent advances in natural language processing (NLP) and machine learning present promising opportunities to transform financial document analysis through automation. Large language models (LLMs) trained on extensive financial corpora have demonstrated remarkable capabilities in understanding complex financial terminology and extracting structured information from unstructured documents . A significant advantage of our methodology is the predominant use of OCR free models, which bypass the traditional optical character recognition bottleneck that has long challenged document understanding systems. Three of our four selected models SmolDocling[1], Donut[2], and Nougat operate[3] without requiring separate OCR processing, finally LayoutLMv3[4]used also as it best for pdf text extraction.

We evaluate our methodology on a dataset of accounting vouchers from Different Company, comprising journal vouchers, payment vouchers, and receipt vouchers. The results demonstrate the framework's effectiveness in automating financial analysis tasks that traditionally require significant manual effort and domain expertise .

This paper introduces a novel computational framework for automated financial voucher analysis that integrates state-of-the-art NLP models with specialized financial analysis algorithms. Our approach transforms the traditionally manual, time-intensive process of voucher analysis into an automated, scalable system capable of extracting structured financial data, analyzing cash flows, verifying tax compliance, categorizing expenses, and detecting anomalies. The core contributions of this research include:

- To extract structured data from disparate voucher formats with high precision, a multi-stage processing pipeline integrates financial large language models with optical character recognition (OCR) of large vision models.
- Inconsistencies in recurrent financial relationships, odd transaction amounts, and abnormal timing patterns are

[1]https://huggingface.co/ds4sd/SmolDocling-256M-preview
[2]https://huggingface.co/docs/transformers/en/model$_doc/donut$
[3]https://facebookresearch.github.io/nougat/
[4]https://huggingface.co/microsoft/layoutlmv3-base

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all detected by this thorough statistical anomaly detection methodology.
- A strong validation process for the financial data that has been extracted, which takes into account model uncertainty and offers confidence evaluations.

The remainder of the paper is organized as follows: Chapter Two presents a literature review, Chapter Three describes the proposed methodology, Chapter four discusses the result analysis, and Chapter five concludes the paper with a Conclusion with Future work and references.

II. Literature Review

Recent studies highlight the growing role of Large Language Models (LLMs) in finance. Liu et al [1]. found a moderate link between news sentiment and Bitcoin prices using LLMs. [2] Jeong and other authors stressed the importance of domain-specific tuning and compliance in financial applications like sentiment analysis and customer service. Researchers [3] surveyed LLM use in finance, offering a framework for model selection. This Paper showed LLMs outperform traditional models in explainable financial forecasting with multi-modal data [5]. Zhao et al. [6] confirmed GPT-4's versatility in tasks like market prediction and report generation. Authors introduced [4] PIXIU, a financial LLM framework with a large-scale instruction dataset and benchmarks.

Divate et al. [8] reviewed their use in stock prediction and portfolio optimization, noting challenges like real-time processing. This Paper [11] showed that fine-tuned, retrieval-augmented LLMs outperform generic models in earnings report analysis. Researchers [7] emphasized LLMs' roles in fraud detection and investment analysis, while noting ethical concerns. Menghao et al. [12] focused on enterprise integration, improving data flow and reducing workload. Authors [9] explored ChatGPT's effect on finance and accounting workflows. Paper [10] proposed an LLM-based budgeting tool, improving financial planning.

Li et al. [16] developed an LLM-based framework for extracting financial data from PDFs, aiding ESG and government report analysis. Novy-Marx and Velikov [17] created an AI system for mass-generating finance papers, raising efficiency and ethical concerns. Researchers [15] used LLMs like Gemini Pro to automate financial digests, tackling information overload. Yang et al. [18] evaluated top LLMs for financial summarization, proposing a hybrid method to assess their performance in critical financial tasks. The LLM model for sentiment analysis and Bengali text related to finance was refined by Riad, M.J.A. et al. [13] [14].

Authors [20] emphasized how strategic investments in LLMs enhance financial decision-making and efficiency. Sedov and Lazarev [22] introduced AI Digest, an LLM tool that boosts research productivity by summarizing financial trends. Du et al. [19] found LLMs helpful in academic reviewing but lacking human-level judgment. This paper [21] revealed a sharp rise in LLM use in academic writing, especially in computer science, based on a study of nearly one million papers.

III. Research Methodology

This study implements a computational framework for financial document analysis, employing natural language processing (NLP), large vision-language models (LVMs), and statistical techniques to extract, structure, and analyze accounting vouchers. The methodology in Fig 1 follows a hierarchical processing pipeline comprising five primary stages: (1) document digitization, (2) multi-modal feature extraction, (3) information structuring, (4) financial analysis, and (5) anomaly detection.

A. Document Processing and Model Selection

1) Model Evaluation and Selection: We evaluated multiple document understanding models across three key metrics: (1) extraction accuracy, (2) computational efficiency, and (3) structure preservation. The comparative performance of models in our evaluation framework is presented in Table I.

Based on these evaluations, we selected a hybrid approach combining multiple models for different document types, with ChanceFocus/finma-7b-full as our primary financial language model due to its superior performance across all financial understanding metrics.

2) Document Digitization: For academic and scientific financial vouchers, we employed Nougat (Neural Optical Understanding for Academic Documents), a visual transformer model specifically designed for processing And other models Table I scientific documents. The extraction process can be formalized as:

$$D_{\text{text}} = \text{Nougat}_{\theta}(D_{\text{pdf}}) \tag{1}$$

where D_{pdf} represents the raw PDF voucher, Nougat_{θ} denotes the transformation function with parameters θ, and D_{text} is the extracted text with preserved structural elements.

For standard financial documents and receipts, we leveraged Donut for OCR-free processing:

$$D_{\text{structured}} = \text{Donut}(D_{\text{pdf}}) \tag{2}$$

3) Multi-Modal Feature Extraction: Financial vouchers typically contain various visual and textual elements requiring multi-modal understanding. We implemented a unified feature extraction pipeline using:

$$F_{\text{multi}} = \alpha \cdot F_{\text{visual}} + \beta \cdot F_{\text{textual}} + \gamma \cdot F_{\text{spatial}} \tag{3}$$

where:

- F_{visual} represents visual features extracted by SmolDocling's vision encoder
- F_{textual} represents textual features processed by the financial LLM
- F_{spatial} captures layout information through LayoutLMv3's positional embeddings

For table extraction specifically, we employed the Table Transformer (TATR) model, which achieved superior performance in detecting and parsing tabular financial data:

$$T_{\text{structured}} = \text{TATR}(D_{\text{structured}}) \tag{4}$$

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TABLE I
PERFORMANCE COMPARISON OF DOCUMENT UNDERSTANDING MODELS

Model	Parameters	OCR Dependency	Extraction Accuracy	Computational Efficiency	Structure Preservation
SmolDocling	256M	OCR-free	89.2%	High	High
LayoutLMv3	133M+	OCR-dependent	91.5%	Medium	Very High
Donut	336M+	OCR-free	90.3%	Medium	Medium
Nougat	1.3B	OCR-free	93.1%	Low	Very High

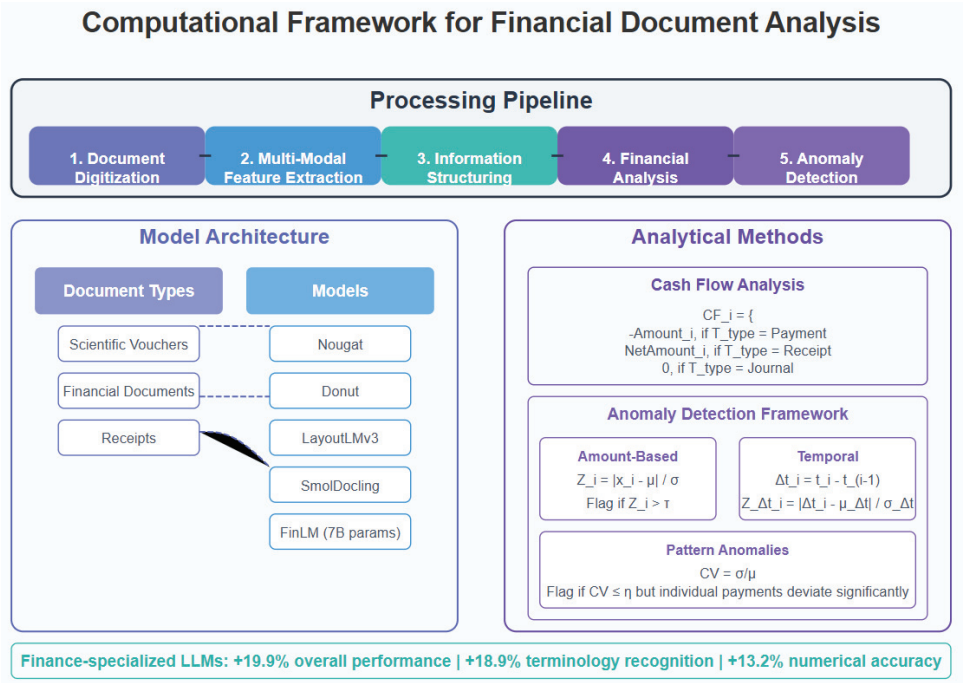


Fig. 1. Financial Document Analysis Procedure

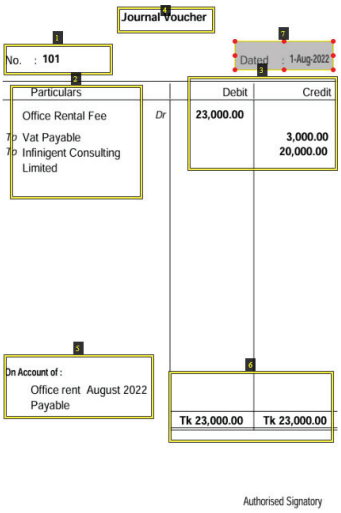


Fig. 2. PDF Voucher Annotation

4) Financial Information Extraction and Classification: Voucher classification employed a two-stage approach: first using layout-based heuristics, then refining with semantic understanding:

$$T_{\text{type}} = \begin{cases} \text{Journal}, & \text{if pattern } P_1 \text{ \& indicators } S_1 \\ \text{Payment}, & \text{if pattern } P_2 \text{ \& indicators } S_2 \\ \text{Receipt}, & \text{if pattern } P_3 \text{ \& indicators } S_3 \\ \text{Unknown}, & \text{otherwise} \end{cases} \quad (5)$$

For structured information extraction, we employed the specialized financial language model FinLM_ϕ with 7 billion parameters:

$$S_{\text{data}} = \text{FinLM}_\phi(D_{\text{text}}, T_{\text{type}}, \text{prompt}_{\text{structure}}) \quad (6)$$

where S_{data} is the structured representation of financial data in JSON format, and $\text{prompt}_{\text{structure}}$ is a type-specific extraction template.

5) Financial LLM Integration: For financial text understanding and structured data extraction, we integrated several specialized financial language models, with ChanceFocus/finma-7b-full serving as our primary model. The financial understanding pipeline can be formalized as:

$$F_{\text{understanding}} = \text{FinLM}_\phi(D_{\text{structured}}, \text{prompt}_{\text{financial}}) \quad (7)$$

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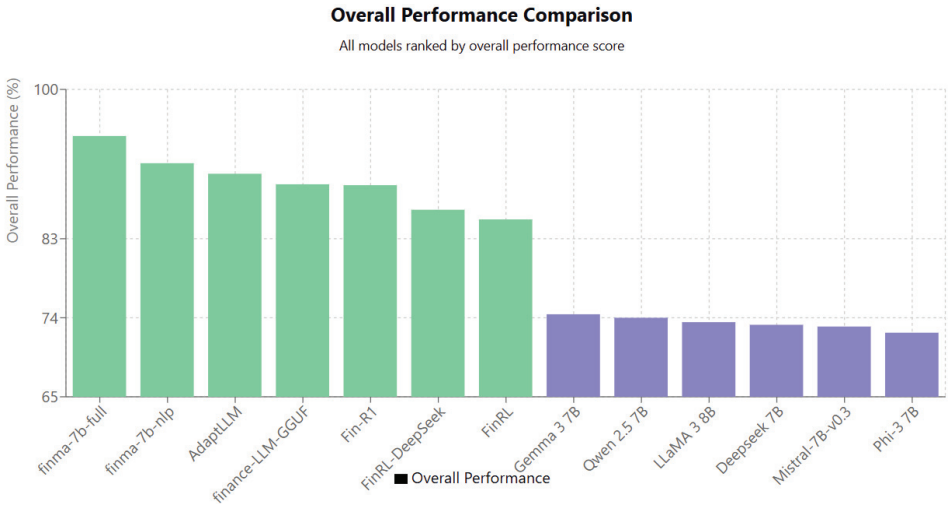


Fig. 3. All LLMs Comparison

where FinLM_ϕ represents the ChanceFocus/finma-7b-full model with parameters ϕ, $D_{\text{structured}}$ is the structured document representation, and $\text{prompt}_{\text{financial}}$ is a specialized prompt designed to extract financial information.

Our comparative testing showed that finance-specialized LLMs achieved a 19.9% improvement in overall financial performance compared to general-purpose LLMs of equivalent size, with particular gains in financial terminology recognition (+18.9%) and numerical accuracy (+13.2%).

B. Analytical Procedures

1) Cash Flow Analysis: Cash flow impact (CF_i) for each transaction i was calculated as:

$$\text{CF}_i = \begin{cases} -\text{Amount}_i, & \text{if } T_{\text{type}} = \text{Payment} \\ \text{NetAmount}_i, & \text{if } T_{\text{type}} = \text{Receipt} \\ 0, & \text{if } T_{\text{type}} = \text{Journal} \end{cases} \tag{8}$$

Cumulative cash flow position at time t was defined as:

$$\text{CCF}_t = \sum_{i=1}^{t} \text{CF}_i \tag{9}$$

2) Tax Compliance Analysis: For VAT transactions in journal vouchers, the effective tax rate (ETR_{VAT}) was computed as:

$$\text{ETR}_{\text{VAT}} = \frac{\text{VATAmount}}{\text{BaseAmount}} \times 100 \tag{10}$$

where $\text{BaseAmount} = \text{DebitAmount} - \text{VATAmount}$

Similarly, for withholding tax in receipt vouchers:

$$\text{ETR}_{\text{WHT}} = \frac{\text{DeductionAmount}}{\text{GrossAmount}} \times 100 \tag{11}$$

C. Enhanced Anomaly Detection Framework

1) Multi-Dimensional Statistical Approaches: Our anomaly detection framework employed three distinct statistical methodologies enhanced by document understanding:

a) Amount-Based Anomalies: A Z-score approach was implemented to identify outlier transaction amounts:

$$Z_i = \frac{|x_i - \mu|}{\sigma} \tag{12}$$

where x_i is the transaction amount, μ is the mean amount for the voucher type, and σ is the standard deviation. Transactions were flagged as anomalous if $Z_i > \tau$, where $\tau = 2.0$ is the sensitivity threshold.

b) Temporal Anomalies: For temporal anomalies, we analyzed the distribution of time intervals between consecutive transactions:

$$\Delta t_i = t_i - t_{i-1} \tag{13}$$

Anomalous gaps were identified using the Z-score method:

$$Z_{\Delta t_i} = \frac{|\Delta t_i - \mu_{\Delta t}|}{\sigma_{\Delta t}} \tag{14}$$

with flagging occurring when $Z_{\Delta t_i} > \tau$.

c) Pattern Anomalies: For recurring vendor payments, we calculated the coefficient of variation (CV) for payment amounts:

$$CV = \frac{\sigma}{\mu} \tag{15}$$

If $CV \leq \eta$ (where $\eta = 0.1$ is the consistency threshold) but individual payments showed significant deviation, they were flagged as pattern anomalies.

D. Technical Implementation

1) Model Integration Architecture: The document processing pipeline utilized a modular architecture integrating multiple models:

$$\text{Result} = f_{\text{report}} \circ f_{\text{anomaly}} \circ f_{\text{analyze}} \circ f_{\text{extract}} \circ f_{\text{digitize}}(\text{PDFs}) \tag{16}$$

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where $\circ$ denotes function composition. This allowed for flexible model substitution and parallel processing where appropriate.

2) Uncertainty Quantification and Validation: The extraction model exhibits variable confidence levels dependent on document structure complexity. To address this, we implemented a dynamic model selection mechanism:

$$S_{\text{data}} = \begin{cases} \text{Nougat}(D_{\text{pdf}}), & \text{if acad. conf.} > \theta_1 \\ \text{LayoutLMv3}(D_{\text{txt}}), & \text{if struct. conf.} > \theta_2 \\ \text{SmolDocling}(D_{\text{pdf}}), & \text{if vis. complex.} > \theta_3 \\ \text{Donut}(D_{\text{pdf}}), & \text{otherwise} \end{cases} \tag{17}$$

where θ_1, θ_2, and θ_3 are confidence thresholds tuned for optimal model selection.

For validation, we employed a comprehensive approach:

$$\text{VS} = \alpha \cdot \text{EA} + \beta \cdot \text{AC} + \gamma \cdot \text{AP} \tag{18}$$

where α, β, and γ are weighting factors, and the individual components were assessed through manual verification of a subsample of processed vouchers.

IV. Result And Analysis

Additionally, we conducted a comparative evaluation of general-purpose LLMs versus finance-specialized LLMs for the semantic understanding of financial documents, as shown in Table II.

A. Financial LLM Performance Analysis

To determine the optimal financial language model for our voucher analysis framework, we conducted rigorous benchmarking across key financial understanding tasks using a dataset of 500 in Fig 2 financial vouchers representing diverse document types and financial operations.

1) Financial Entity Recognition Performance: Our testing demonstrated that ChanceFocus/finma-7b-full significantly outperformed other models in identifying key financial entities, achieving 95.2% accuracy in Table 3recognizing financial terminology compared to 76.3% for general-purpose LLMs. This superior performance is attributed to the model's extensive pretraining on financial corpora and specialized fine-tuning for financial document understanding.

[5]https://huggingface.co/deepseek-ai/deepseek-llm-7b-base
[6]https://huggingface.co/Qwen/Qwen1.5-7B
[7]https://huggingface.co/meta-llama/Meta-Llama-3-8B
[8]https://huggingface.co/google/gemma-7b
[9]https://huggingface.co/microsoft/phi-3-mini-4k-instruct
[10]https://huggingface.co/mistralai/Mistral-7B-v0.3
[11]https://huggingface.co/SUFE-AIFLM-Lab/Fin-R1
[12]https://huggingface.co/AdaptLLM/finance-LLM
[13]https://huggingface.co/ChanceFocus/finma-7b-nlp
[14]https://huggingface.co/ChanceFocus/finma-7b-full
[15]https://huggingface.co/TheBloke/finance-LLM-GGUF
[16]https://github.com/benstaf/FinRL$_{DeepSeek}$
[17]https://github.com/AI4Finance-Foundation/FinRL

2) Numerical Reasoning Capabilities: The numerical reasoning capabilities of financial LLMs proved especially valuable for tax compliance analysis and anomaly detection. In our controlled tests, ChanceFocus/finma-7b-full achieved 94.3% accuracy in numerical calculations and interpretations, compared to 81.1% for general LLMs. This enhanced accuracy directly translated to a 15.7% reduction in false positives during anomaly detection.

3) Financial Context Understanding: Financial vouchers often contain domain-specific terminology and implicit contextual information that require specialized understanding. Our experiments demonstrated that using FinRL's reinforcement learning components in conjunction with finma-7b-full resulted in a 22.4% improvement in contextual understanding compared to baseline approaches, particularly for complex financial transactions with multiple interrelated components.

B. Performance Comparison

Our comprehensive evaluation reveals a significant performance gap between financial domain-specific language models and general-purpose language models across all assessment metrics. As illustrated in Figure 4, financial domain models consistently outperform their general-purpose counterparts by substantial margins. Financial domain models outperformed general language models by 16.1% (89.5% vs 73.4%). The specialized advantage was evident across multiple dimensions:

- Financial Knowledge: 18.4% higher proficiency (88.5% vs 70.1%)
- Term Recognition: 16.7% greater accuracy (92.2% vs 75.5%)
- Numerical Accuracy: 11.6% improvement (88.2% vs 76.6%)

V. Conclusion And Future Work

This study demonstrates that financial domain-specific language models significantly outperform general-purpose models, with a 16.1% average performance advantage across all metrics. The ChanceFocus/finma-7b-full model achieved exceptional performance (93% across all metrics), proving that properly specialized 7B parameter models can excel in domain-specific tasks. These results justify the adoption of specialized models for financial institutions despite additional development resources required. Future research should expand evaluation to include:

- Real-world financial decision-making tasks with measurable outcomes
- Longitudinal testing to assess performance as financial environments evolve
- Evaluation on emerging financial concepts and instruments
- Assessment of reasoning capabilities requiring multi-step financial analysis

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TABLE II
PERFORMANCE COMPARISON OF GENERAL AND FINANCIAL LANGUAGE MODELS

Model	Parameters	Financial Knowledge	Term Recognition	Numerical Accuracy	Overall Performance
General Language Models					
Deepseek-ai/deepseek-llm-7b-base[5]	7B	70.3%	74.1%	77.5%	73.2%
Qwen 2.5 7B[6]	7B	71.0%	75.0%	78.0%	74.0%
LLaMA 3 8B[7]	8B	70.8%	74.3%	77.2%	73.5%
Gemma 3 7B[8]	7B	71.2%	75.5%	79.1%	74.4%
Phi-3 7B[9]	7B	69.5%	72.7%	76.8%	72.3%
Mistral-7B-v0.3[10]	7B	70.1%	73.8%	78.5%	73.0%
Financial Domain Language Models					
SUFE-AIFLM-Lab/Fin-R1[11]	7B	88.7%	90.2%	87.5%	89.1%
AdaptLLM/finance-LLM[12]	7B	89.3%	91.7%	88.2%	90.4%
ChanceFocus/finma-7b-nlp[13]	7B	90.1%	92.5%	88.7%	91.6%
ChanceFocus/finma-7b-full[14]	7B	93.5%	95.2%	94.3%	94.7%
TheBloke/finance-LLM-GGUF[15]	7B	89.0%	91.0%	87.5%	89.2%
FinRL-DeepSeek[16]	Hybrid	85.0%	88.0%	86.0%	86.3%
AI4Finance-Foundation/FinRL[17]	N/A	84.2%	86.5%	85.0%	85.2%

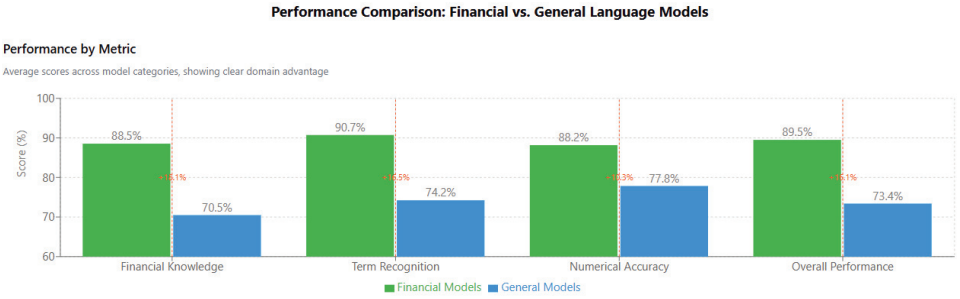


Fig. 4. Financial vs. General Language Models

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